Precision Actuation of a Ball-and-Socket Joint Using Shape Memory Alloy Springs

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Brajesh Budhathoki, Yiunfan Hu, Jonathan Jiang

Faculty Advisors: Dr. Nilanjan Chakraborty, Dr. William Stewart & Dr. Shanshan Yao

Department of Mechanical Engineering, Stony Brook University, Stony Brook, NY, 11794, USA

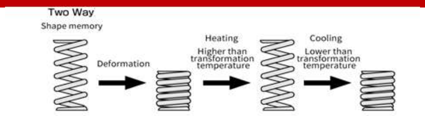
Stony Brook Center for Inclusive Education

Abstract

This study presents the development and control of a novel multiple degree of freedom (DOF) ball and socket joint actuated by shape memory alloy (SMA) springs. The proposed joint aims to enhance the flexibility and functionality of robotic systems, particularly in applications requiring precise and adaptive movements. The design integrates SMAs within the ball and socket mechanism to achieve controlled, reversible motion across multiple axes. We aim to emulate the opposable thumb, shoulder and wrist of humans. SMA-controlled joints provide a promising solution for advanced robotic applications, offering improved maneuverability and adaptability compared to traditional rigid joints.

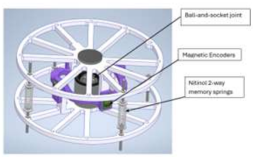
Background

Modern robotic systems often require joints capable of multi-degree-of-freedom movement to mimic the complex motions found in biological systems. However, traditional joints (e.g. motors, hydraulic systems, mechanical actuators) face several limitations in complexity, bulkiness, noise and energy consumption. The implementation of Shape Memory Alloys (SMAs) provides a novel alternative to traditional joints. SMAs are a unique group of metallic materials known for their ability to return to a pre-defined shape when subjected to the appropriate thermal procedure. This property is due to a reversible phase transformation between martensite and austenite. Given these unique properties, SMAs prove to be an excellent option for a compact, flexible and energy efficient robotic joint.

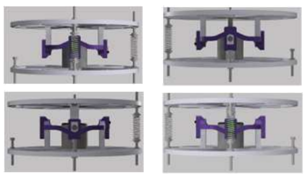


Methods and Materials

Design Overview:



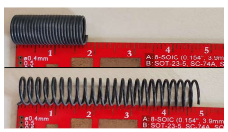
CAD assembly of the whole system



Front, back, left, and right view

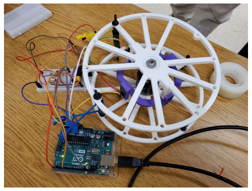
Materials Used:

- SMA Springs: Nitinol 2-way memory coil springs.



- Operation: Expands at 60°C, contracts at room temperature, providing force in both directions.
- Specifications: Wire gauge: 0.7 mm, Diameter: 9 mm, Length (cold): 14 mm, Length (hot): 85 mm, Weight: 1.28 g.

- Platforms: 3D-printed PLA, used to mount and position the SMA springs.
- Control Electronics: MOSFETs (used as a switch), Magnetic Encoders (position feedback), Arduino (controls MOSFETs & encoders).



Control Circuit Layout

System Overview:

Experimental Setup

- Assembly: SMA springs at 90 ˚ for 2-axis control.
- Control circuit: MOSFETs and power supply.

Operation

- Inputs: Enter target angles Θ_x and Θ_y.
- Encoders: Read and adjust values and convert them to angles.

Rotation and Displacement

- Initial setup: Identify initial attachment points $\mathbf{P}_{\mathbf{initial}}$.
- Apply rotation:

$$R_x(\Theta_x): \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\theta_x & -sin\theta_x \\ 0 & sin\theta_x & cos\theta_x \end{bmatrix} \qquad R_y(\Theta_y): \begin{bmatrix} cos\theta_y & 0 & sin\theta_y \\ 0 & 1 & 0 \\ -sin\theta_y & 0 & cos\theta_y \end{bmatrix}$$

Combined:

$$R = R_y . R_x$$

New position: $P_{rot} = R.P_{initial}$

Length Change

- $\Delta L_1 = Length_{rotated} - Length_{original}$; $\Delta L_2 = Length_{rotated} - Length_{original}$

Control

- Adjust: SMA springs via MOSFET based on ΔL

Results

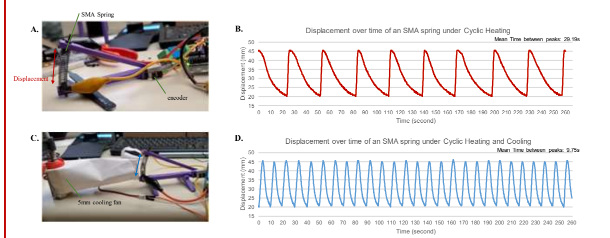


Figure 1. Cycle times of SMA Spring with and without cooling. A) One-axis simplification of the ball joint used to gather these data. An absolute magnetic encoder is used to calculate displacement. B) Results from cycling SMA spring from minimum to maximum displacement at an ambient temperature of 27°C. C) Experimental set-up with a 5mm cooling fan and paper cone directing airflow over the spring. D) Results with the fan activated to assist spring cooling.

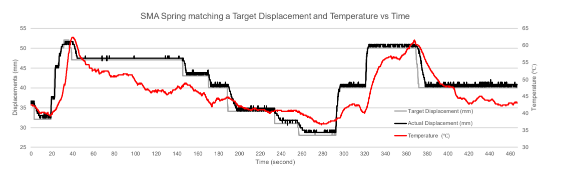


Figure 2. Tracing a target curve using a displacement-based threshold for heating and cooling. This is close to the control scheme for the final ball and socket joint.

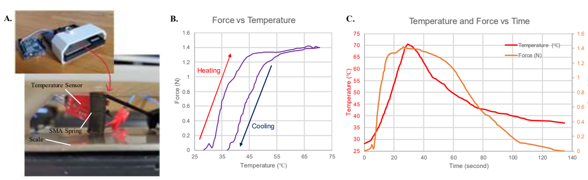
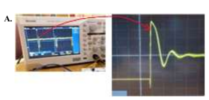


Figure 3. Relationship between force and temperature. A) Force measurement set-up using a scale with the spring pushing down. B) Correlation between force and temperature shows a clear hysteresis. C) The data presented in B plotted with a time axis



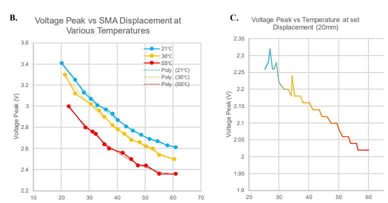


Figure 4. Investigating relationship between transient voltage peak, displacement, and temperature. A) Using MOSFETS to provide PWM signal revealed a spike in voltage at the beginning of each cycle. B) Voltage peak and displacement with the spring at 3 constant temperatures with a mechanically induced displacement. C) Voltage peak and temperature at a constant displacement. Note the y-axis scale is different as the input voltage between the experiments are different.

Summary

The experiment setup successfully demonstrates the shape memory effect of the SMA spring, highlighting its contraction upon cooling and expansion upon heating. The cycle times between heating and cooling shown in figure 1 displays the periodic relationship between the displacement due to heating and cooling against time. This characteristic is pivotal for precise control of a ball-and-socket joint system, as the SMA springs' responsive nature allows for accurate adjustments of the joint's orientation. The results from this setup provide valuable insights into the SMA springs' behavior, guiding the optimization of temperature control and actuation strategies for enhanced system performance as we can see from figures 2 and 3 that the displacement of the SMA springs can be determined through measurable data such as force and temperature. By carefully regulating the heating and cooling via MOSFETs and Arduino (shown in figure 4), accurate and responsive movements were achieved along two axes, using two SMA springs placed 90˚ apart. The inclusion of the magnetic encoders provided precise feedback, ensuring stable and controlled adjustments. These findings highlights the importance of temperature control and feedback mechanisms in optimizing the performance of SMA-driven actuation systems.

Future Directions

- Draw further inspiration from the human anatomy to create a more compact and reliable prototype of the robotic joint.
- Determine and model a relationship between inductance and SMA spring displacement to implement a sensorless design of the robotic joint.
- Introduce a third degree of freedom with SMA-based torsion actuators.

Robotic Joint Prototype inspired by human anatomy

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